Sarah Oh

PhD Candidate, UC Berkeley
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Computational cognitive scientist specializing in human learning and decision-making. Combines behavioral experiments with interpretable cognitive models to identify computational principles underlying the efficiency and flexibility of human behavior, with an interest in applying these methods to probe and characterize complex behaviors in AI systems.

EDUCATION

University of California, Berkeley – Berkeley, CA

- **Ph.D.** in Psychology, cognitive science emphasis — 2022 – present

Dartmouth College – Hanover, NH

- **M.S.** in Engineering Sciences — 2017 – 2019
- **B.A.** in Engineering Sciences modified with Computer Science – *cum laude* — 2013 – 2017

SKILLS

Programming: Python, R, MATLAB, JavaScript, Stan, Java, C, Bash, Git
Machine learning: reinforcement learning, PyTorch, TensorFlow, scikit-learn
Research: experimental design, computational cognitive modeling, Bayesian inference, simulation-based inference

RESEARCH & WORK EXPERIENCE

University of California, Berkeley – Berkeley, CA
Graduate Student Researcher — August 2022 – present

- **Cognitive mechanisms of habit formation and cognitive control**
  - Designed a multi-step decision-making paradigm in JavaScript to accelerate habit induction in human participants and used hierarchical Bayesian models to assess evidence for habit formation, resulting in a first-author publication at JEP:G.
  - Develop reinforcement-learning-based computational models to expose mechanisms underlying habit formation.
  - Validate the experimental paradigm against psychometric self-report measures of real-world habit tendencies.
- **Computational models of cognitive flexibility in humans and animals**
  - Develop computational models of human and mouse behavior in a multimodal rule learning task.
  - Train RNNs in PyTorch to perform simulation-based inference over generative model class, model parameters, and latent cognitive variables.

California Institute of Technology – Pasadena, CA
Research Technician/Assistant — November 2019 – July 2022

- **Neural substrates of goal-directed vs. habitual control**
  - Developed behavioral tasks to probe model-based/model-free reinforcement learning and habit formation during fMRI.
  - Collected fMRI data and set up preprocessing and statistical modeling pipelines in SPM.
- **Large-scale computational phenotyping for computational psychiatry**
  - Implemented and deployed 4 cloud-hosted decision-making tasks via jsPsych and Heroku.
  - Developed computational models to relate behavioral parameters with psychiatric symptom dimensions.
- Administered the lab's website, data storage, and computing servers.

Dartmouth College – Hanover, NH
Master's Student — Fall 2017 – Spring 2019

- Thesis: ***Towards a Perceptual Distance Metric for Auditory Stimuli***
  - Modeled transformations applied to stimuli during early auditory processing as a function mapping points in stimulus space to perceptual space, inducing a Riemannian distance metric validated against human-rated subjective dissimilarity between reference and degraded audio clips.

MIT Lincoln Laboratory – Lexington, MA
Summer Research Program Intern

- **Automated network protocol characterization** — Summer 2017
  - Researched and applied methods from computational mechanics and machine learning to network traces for unsupervised characterization of network protocols.
- **Embedded system interrogation and profiling** — Summer 2016
  - Developed a lightweight, modular shell utility in C/C++ to enable interrogation of unknown embedded systems.

PUBLICATIONS

- ***Oh, S.**, Collins, A. G. E. "Naturally disengaging control to reveal habits." *J. Exp. Psychol. Gen.* (in press).
- *Cockburn, J., Ding, W., **Oh, S.**, Charpentier, C., Tadayonnejad, R., Feusner, J. D., O'Doherty, J. P. "Characterizing heterogeneity in human reinforcement learning and the arbitration of behavioral control." *Nat. Commun.* (in press).
- *Johri, A., Kluen, L., Gera, R., Man, V., Perez, O., Simon, J. P., Ding, W., Fransen, A., Cho, **S., Oh**, S., Cockburn, J., Feusner, J. D., Tadayonnejad, R., & O'Doherty, J. P. "Distributed activity in the human posterior putamen distinguishes goal-directed from habitual control." *Curr. Biol.* (in press).
- Bähner, F., **Oh, S.**, Hummos, A., Sandbrink, K., Meyer-Lindenberg, A., & Collins, A. G. E. "Neurocomputational mechanisms underlying cognitive flexibility: A translational perspective." Trends Cogn. Sci. (2026).
- *Ding, W., Cockburn, J., Simon, J. P., Johri, A., Cho, S. J., **Oh, S.**, Feusner, J. D., Tadayonnejad, R., O'Doherty, J. P. "Model-based and model-free valuation signals in the human brain vary markedly in their relationship to individual differences in behavioral control." *Cell Rep.*, 45, 117454 (2026).
- *Perez, O. D., ***Oh, S.**, Dickinson, A., Arenas, J., & Merlo, E. "Human goal-directed behavior is resistant to interventions on the action-outcome contingency." *BMC Psychology*, 14, 444 (2026).
- ***Oh, S.**, Collins, A. G. E. "Computational insights from a novel habit induction protocol," *Proceedings of the Annual Meeting of the Cognitive Science Society* (2025).
- *Wu, Q., **Oh, S.**, Tadayonnejad, R., Feusner, J. D., Cockburn, J., O'Doherty, J. P., Charpentier, C. J. "Individual differences in autism-like traits are associated with reduced goal emulation in a computational model of observational learning." *Nat. Mental Health* 2, 1032–1044 (2024).

* denotes first-authorship

CONFERENCE PRESENTATIONS

- "Validating an experimental paradigm for inducing and measuring habits." CogSci 2026 (Poster)
- "Computational insights from a novel habit induction protocol." CogSci 2025 (Talk)
- "Exploring the computational mechanisms of habit learning in a novel habit task." RLDM 2025 (Poster)
- "Revealing habits in a new, short hierarchical paradigm." Psychonomic Society 2023 (Poster)

TEACHING EXPERIENCE

- **Graduate Student Instructor**, UC Berkeley
  - COGSCI C127: Cognitive Neuroscience — Spring 2026
  - COGSCI C131: Computational Models of Cognition — Fall 2023

SERVICE & LEADERSHIP

- **Peer review:** NeurIPS Workshop on Intrinsically Motivated Open-ended Learning (IMOL) 2024, CogSci 2025
- **Mentorship**: mentored students through Research Experience Pathways (REP), which provides paid research experiences to underrepresented students at UC Berkeley (2024 – present)